

5737241

Ubuntu Virtual Environment Configuration, Recovery, and Backup Automation using Command-line Tools



[Created using PicLumen: <https://piclumen.com/app/image-generator/create>]

Contents:

Section A –

- ***Question 1 - Virtual Machine and OS Configuration
[50 Marks]***
- ***Question 2 - Troubleshooting Video
[20 Marks]***
- ***Question 3 - Scripting
[10 Marks]***

Question 1: Virtual Machine and OS Configuration

1.1 Install Oracle VM software and create two virtual machines using Oracle VirtualBox.

Installation of Oracle VMware:

1) Download Oracle VMware for a Windows host from the Website.



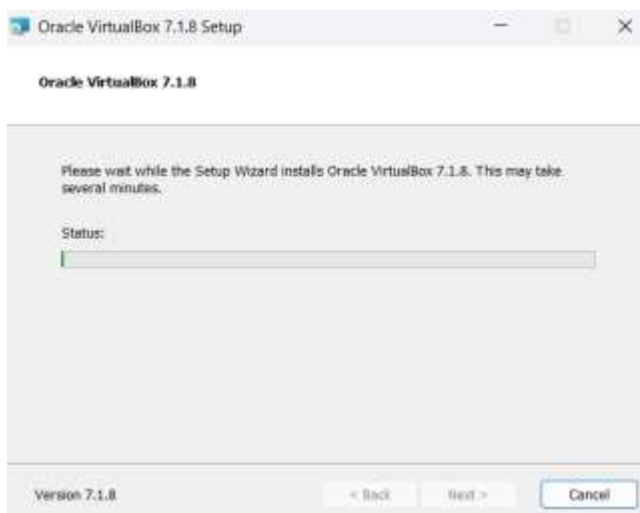
2) Follow the prompts.



3) Open the setup Wizard.



4) Allow it to install.



- **Creating the First Virtual Machine:**

Virtual Machine 1: Linux2030

Allocated Size: 60 GB

Operating System: Ubuntu 24.04

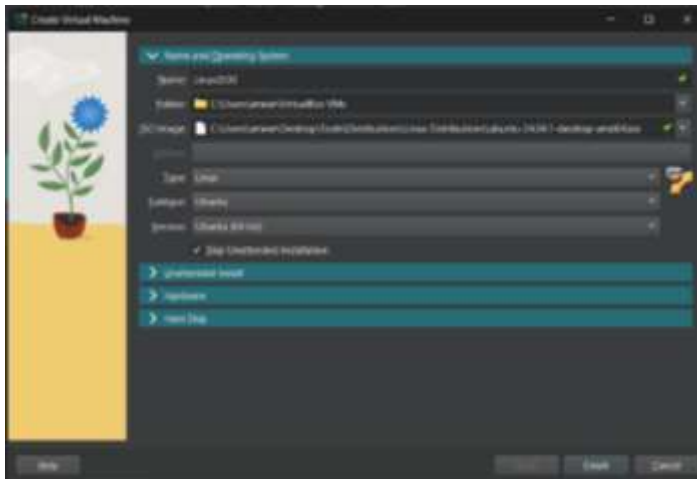
Administrator Username: Ameer

Administrator Password: L1n@x2030

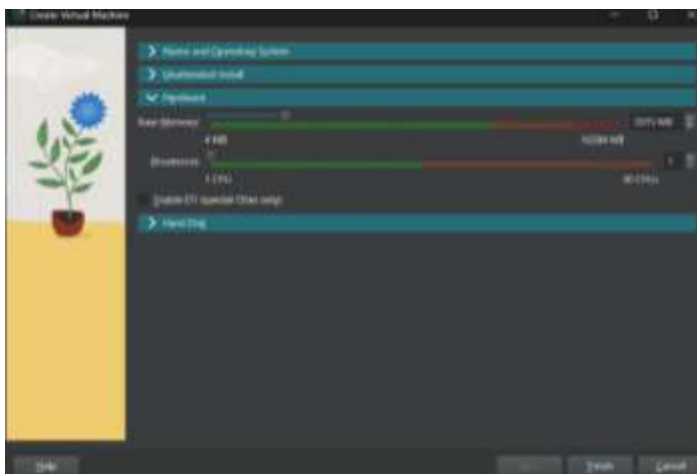
RAM Size: 3GB

Screenshots:

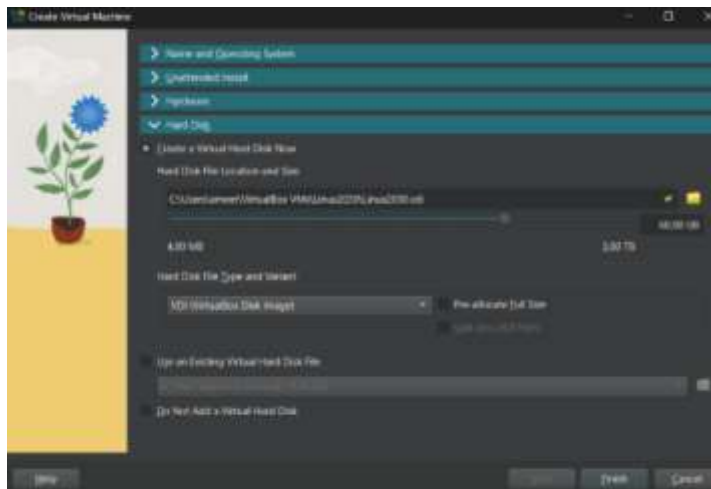
1) Naming the VM “Linux2030”



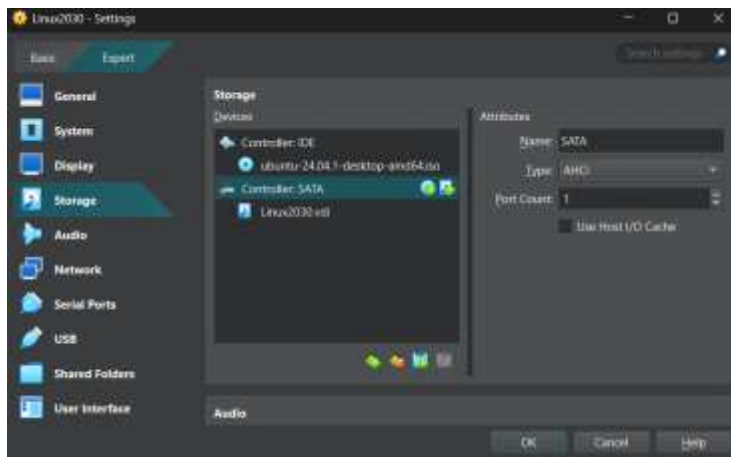
2) Allocated 3GB RAM to the machine.



3) Created the Virtual Disk with 60GB.

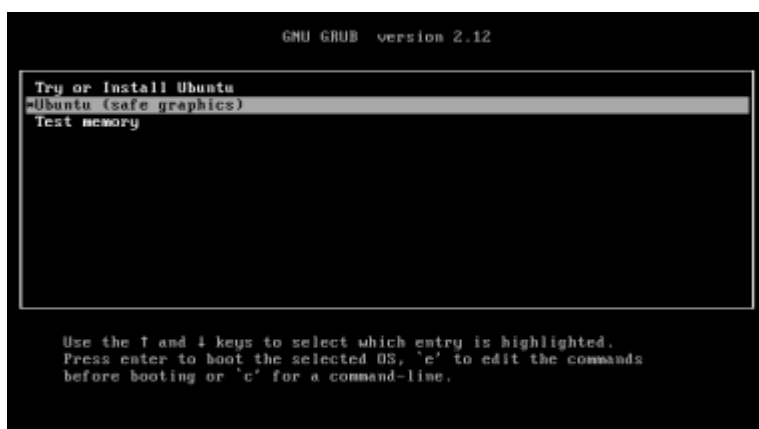


4) Adding the IDE, using the ubuntu 24.04 ISO, to the VM.



- Installing the Ubuntu Operating System on VM1 “Linux2030”:

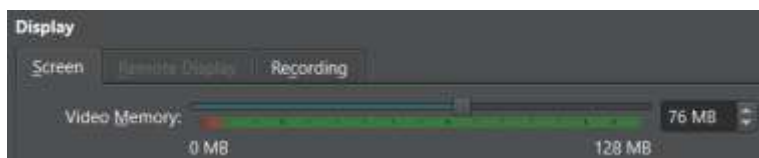
1) Run the VM. Navigate to the second option using the arrow keys. Press Enter key.



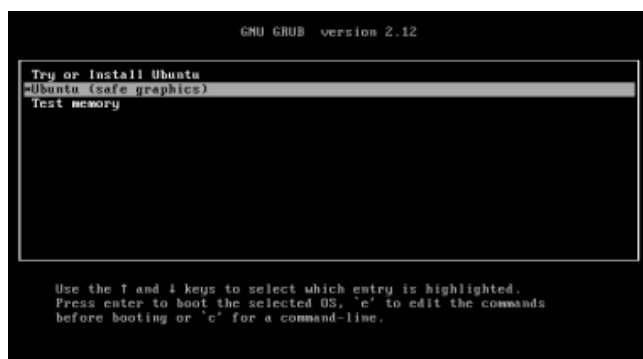
Note: I experienced a crash as Ubuntu was booting up.



- Power off the Vm.
- Navigate to the settings menu in the Oracle VMware application.
- In the settings menu, go to Display. I increased the Video memory that is allocated to the machine to 76MB.



- Start the VM again



2) The Ubuntu installation will begin.



3) Select the interactive Installation.

Type of installation



How would you like to install Ubuntu?

☒ **Interactive installation**
For users who want to be guided step by step through the installation.

☐ **Automated installation**
For advanced users who have an automake1 yami for container and repeatable system setups.

Back Next

4) Continue to Follow the prompts. Select the Options that suit you.

Keyboard layout



Select your keyboard layout

Detected

English (UK)
English (US)
Esperanto

Select your keyboard variant: English (US)

Type here to test your keyboard

Back Next

5) Select the first option.

Disk setup



How do you want to install Ubuntu?

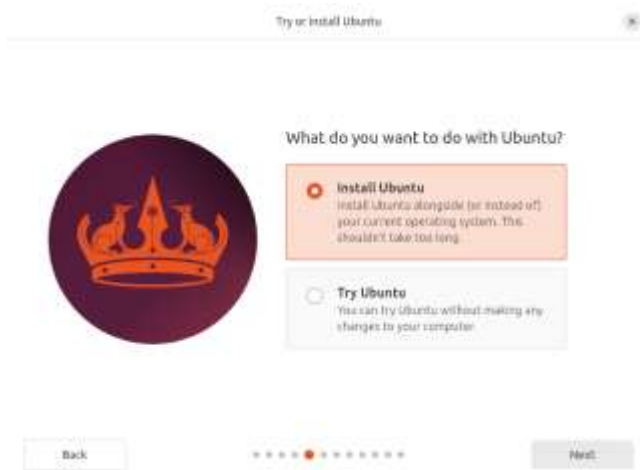
☒ **Erase disk and install Ubuntu**
Start from scratch on your selected disk.

Advanced features... (none selected)

☐ **Manual installation**
For advanced users seeking customized disk setups.

Back Next

6) Choose to Install the operating system.



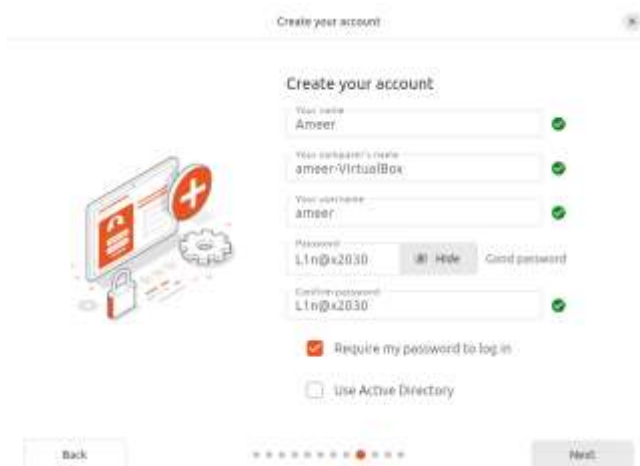
7) Ensure the location is accurate if you're open to it, or disclose your location.



8) Continue to follow the prompts.



9) I populated the details according to the instructions.



Create your account

Create your account

Your name: Ameer ✓

Your computer's name: ameer-VirtualBox ✓

Your username: ameer ✓

Password: L1n@x2030 ✓ Good password

Confirm password: L1n@x2030 ✓

☒ Require my password to log in

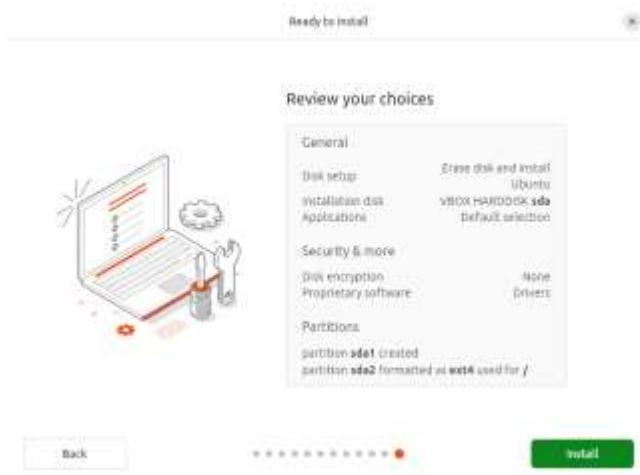
☐ Use Active Directory

Back Next

Note: Name = *Ameer*

Password = *L1n@x2030*

10) Review the choices.



Ready to install

Review your choices

General

Disk setup: Erase disk and install Ubuntu

Installation disk: VBOX HARDISK sda

Applications: Default selection

Security & more

Disk encryption: None

Proprietary software: Drivers

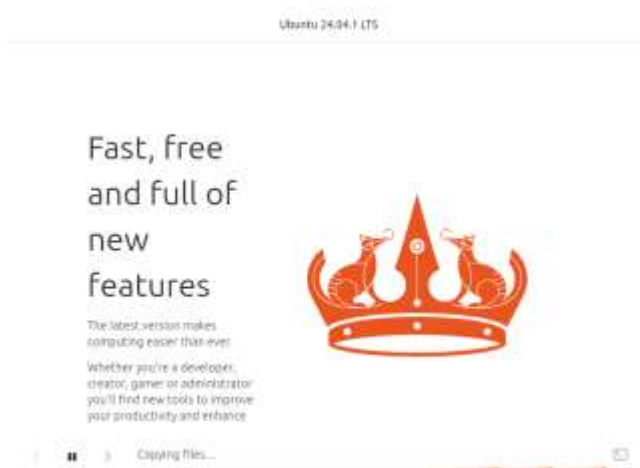
Partitions

partition sda1 created

partition sda2 formatted as ext4 used for /

Back Install

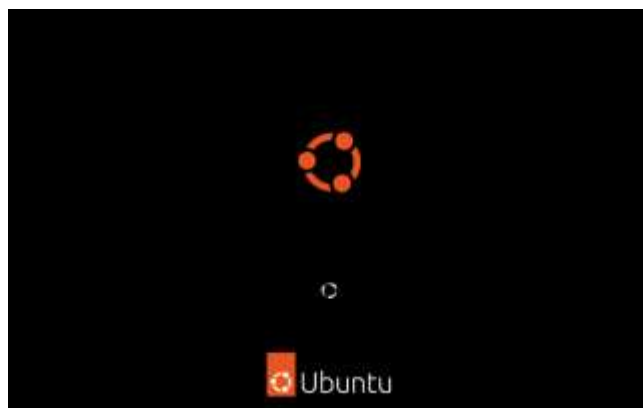
11) Allow Ubuntu to install.



12) It should take around 20-30 minutes.



13) Allow the Vm to restart.



14) The system prompted me to Login to my account.



Note: Password = *L1n@x2030*

15) I was successfully able to launch the Ubuntu OS.

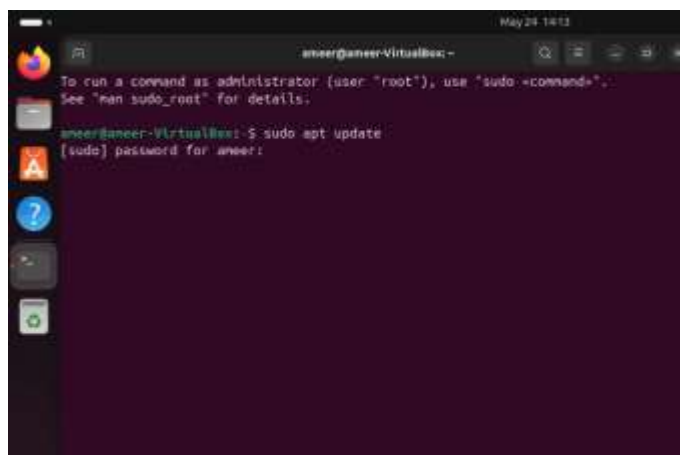


- Enable Full Screen:

1) Open the terminal.

Enter the command:

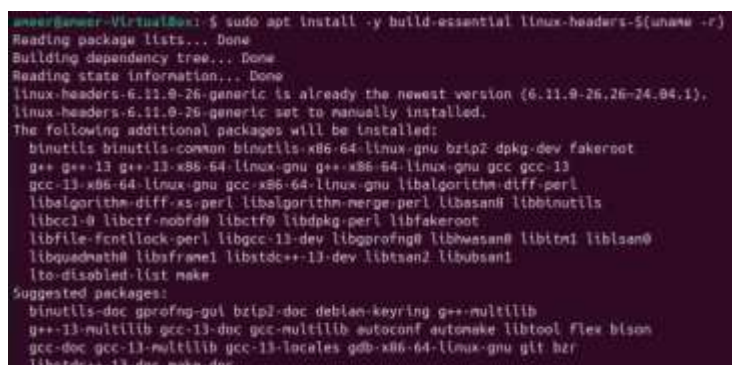
“sudo apt update”



Note: Enter your user password to gain sudo privileges.

2) Next, enter the command:

“sudo apt install -y build-essential linux-headers-\$(uname -r)”



Note: Close the terminal after the download.

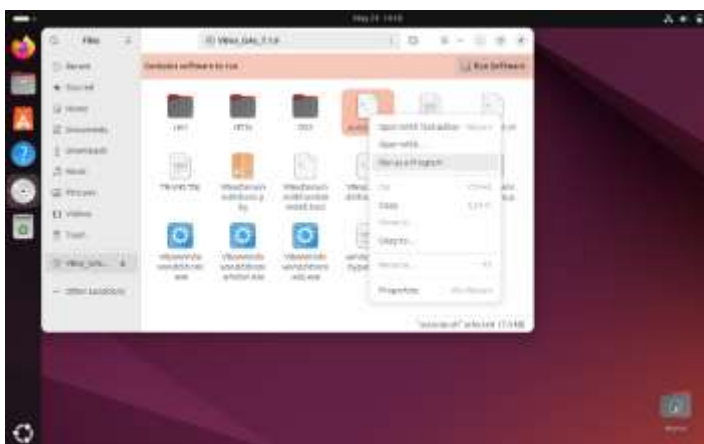
- 3) Navigate to the top left tabs. Select Devices and press 'Insert Guest Addition CD image'**



- 4) Find the CD icon on the left menu column. It is located above the Recycle bin application.**



- 5) Right click on the 'autorun.sh' file. Select the option to run the file as a program.**



Note: It will prompt you to enter your password, please comply.

6) Go to the top right GUI panel and restart the machine.



7) Once the machine is online again, navigate to the top left tabs. Select 'View' and enter Fullscreen mode.



- **Your VM should be able to enter fullscreen mode.**



- **Creating the Second Virtual Machine:**

Virtual Machine 2: PSG2024

Allocated Size: 25 GB

Operating System: Ubuntu 24.04

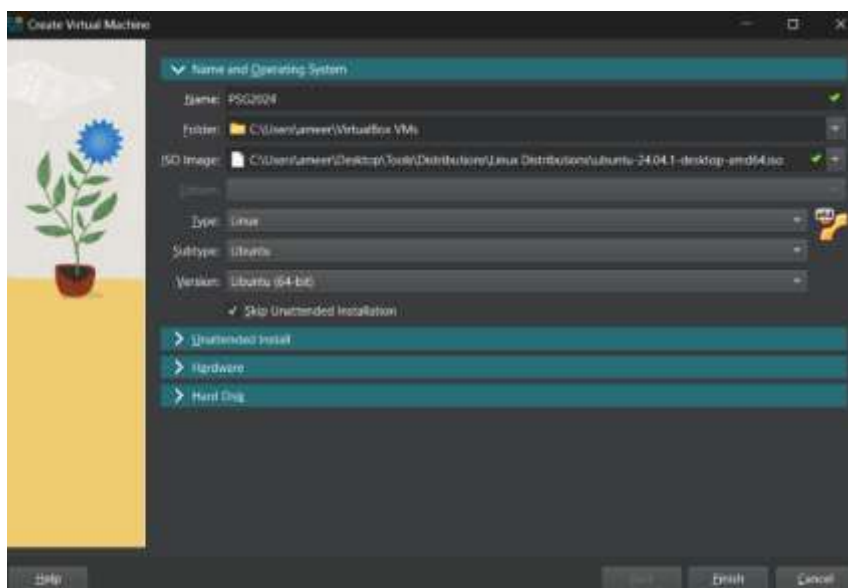
File Partition: NTFS

Administrator Username: Kylian Mbappé

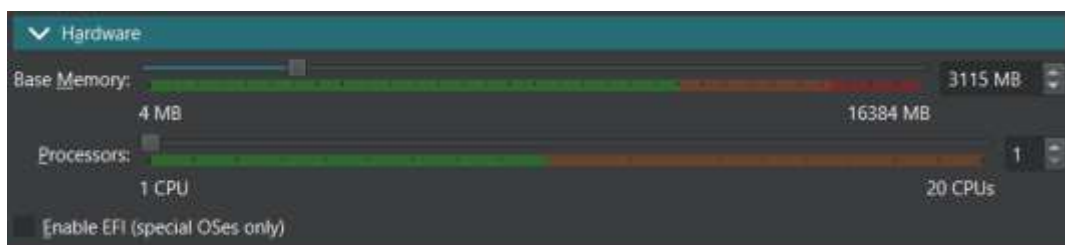
Administrator Password: L1n@x2000

Screenshots:

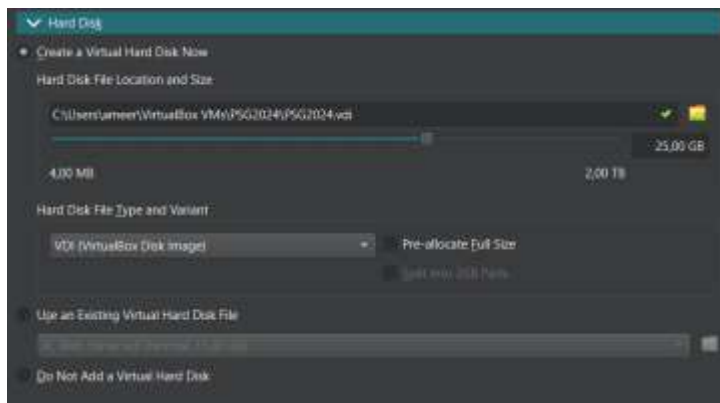
1) Create a new VM using Oracle VMware



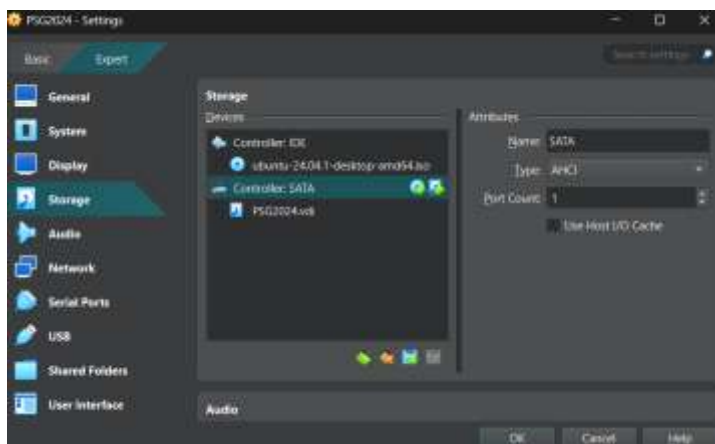
2) Allocated 3GB RAM to the machine.



3) Allocated 25GB of Virtual Disk.



4) Selecting the IDE, using the Ubuntu 24.04 ISO.



Installing the Ubuntu 24.04 ISO On VM2 “PSG2024”:

1) Start the VM.



2) Choose the Interactive Option.

Type of installation

How would you like to install Ubuntu?

 **Interactive installation**
For users who want to be guided step by step through the installation.

☐ **Automated installation**
For advanced users who have an automised yaml for consistent and repeatable system setups.

Back Next

3) Follow the prompts.

Keyboard layout

Select your keyboard layout Detect

English (UK)
English (US)
Esperanto

Select your keyboard variant: English (US)

Type here to test your keyboard

Back Next

4) I populated the details according to the instructions.

Create your account

Create your account

Your name
Kylian Mbappe

Your computer's name
kylian-mbappe-VirtualBox

Your username
kylian-mbappe

Password
L1n@x2000 Hide Good password

Confirm password
L1n@x2000

☒ Require my password to log in

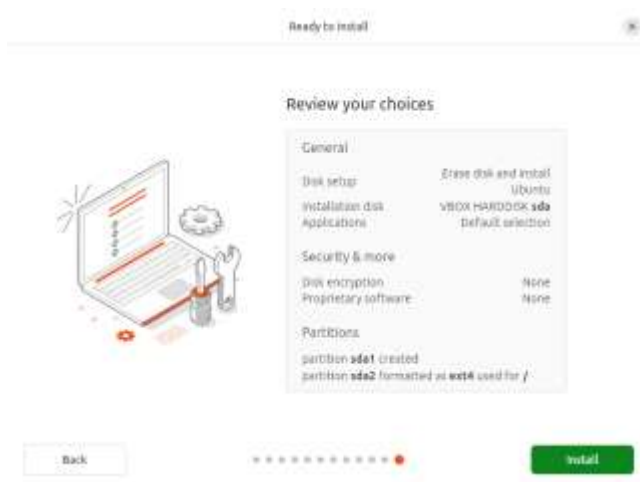
☐ Use Active Directory

Back Next

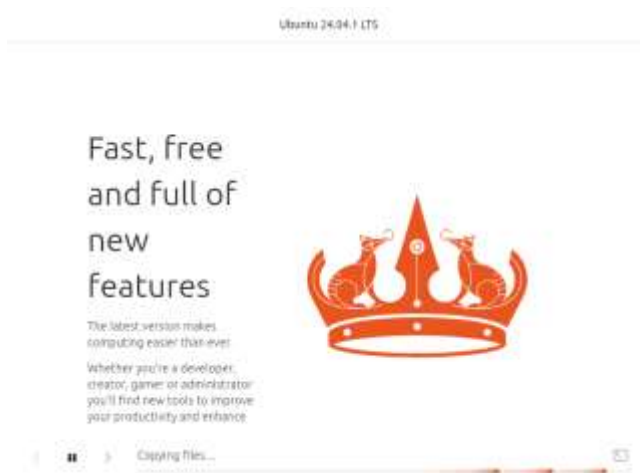
Note: Name = *Kylian Mbappe*

Password = *L1n@x2000*

5) Review.



6) Allow Ubuntu to install.

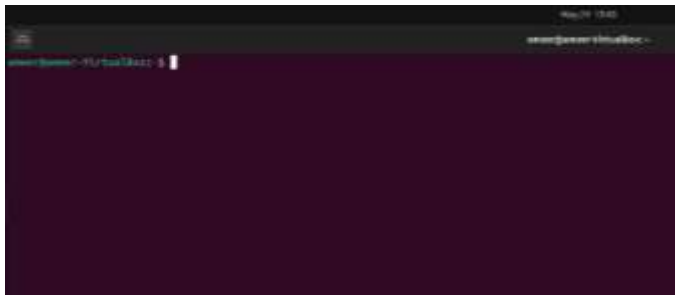


1.2 Create a new sub-directory called Wildlife in your home directory.

Relaunch VM1 'Linux2030':



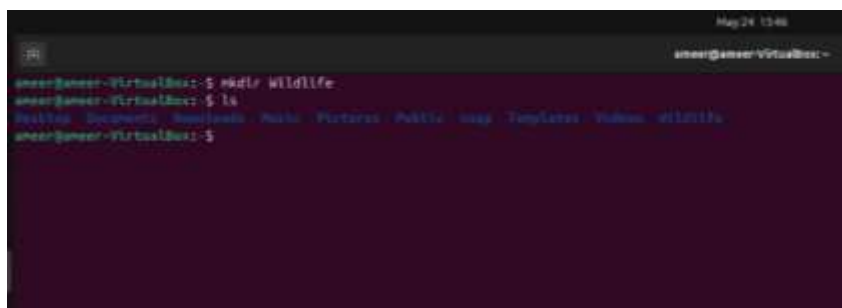
1) Open the terminal.



2) Create the new directory called 'Wildlife'

Command:

"mkdir Wildlife"

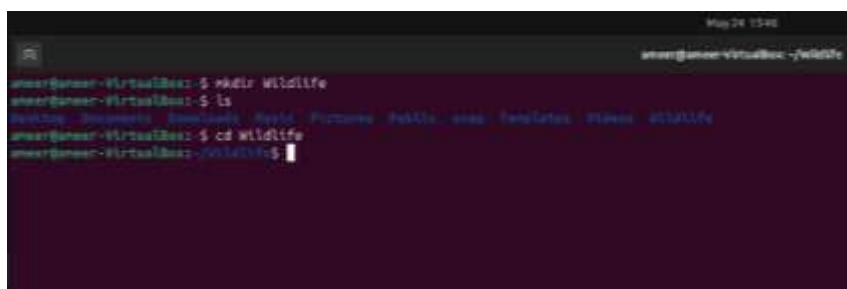


Note: ls = shows directories under the home tab.

3) Navigate to the Wildlife directory:

Command:

"cd Wildlife"



1.3 Create a text file called Namibia in the Wildlife directory.

Command:

“nano Wildlife/Namibia”

1) Enter the given text into the file.

Text:

“Oryx

Springbok

Strandjut

Puku

Sitatunga

Topi

Oribi

Kudu”



```
May 24 15:50
ameer@ameer-VirtualBox: ~/Wildlife
GNU nano 2.2.2 Namibia *
Oryx
Springbok
Strandjut
Puku
Sitatunga
Topi
Oribi
Kudu
^
```

2) Use ‘CTRL + X’ then ‘y’ and press ‘Enter’ to Save and exit.

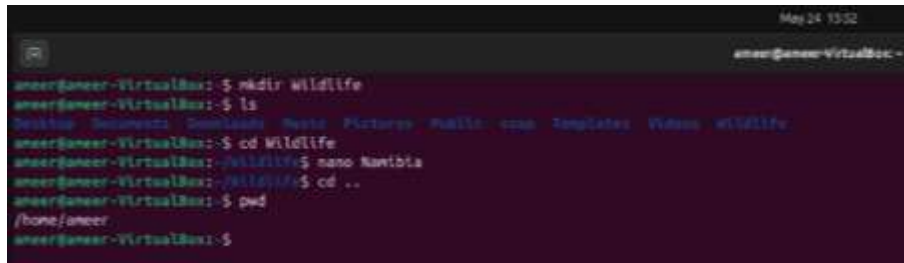


```
May 24 15:51
ameer@ameer-VirtualBox: ~/Wildlife
ameer@ameer-VirtualBox:~$ mkdir Wildlife
ameer@ameer-VirtualBox:~$ ls
Desktop Downloads Downloads Music Pictures Public Templates Videos Wildlife
ameer@ameer-VirtualBox:~$ cd Wildlife
ameer@ameer-VirtualBox:~/Wildlife$ nano Namibia
ameer@ameer-VirtualBox:~/Wildlife$
```

3) Return to the Home directory:

Command:

`"cd .."`



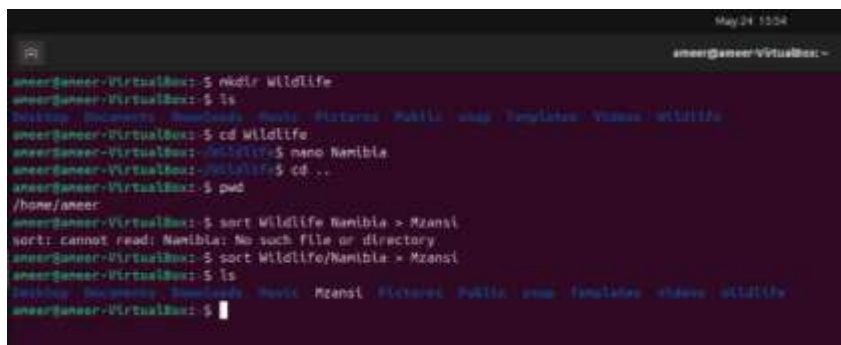
```
ameer@ameer-VirtualBox: ~$ mkdir Wildlife
ameer@ameer-VirtualBox: ~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos Wildlife
ameer@ameer-VirtualBox: ~$ cd Wildlife
ameer@ameer-VirtualBox: ~/Wildlife$ nano Namibia
ameer@ameer-VirtualBox: ~/Wildlife$ cd ..
ameer@ameer-VirtualBox: ~$ pwd
/home/ameer
ameer@ameer-VirtualBox: ~$
```

Note: pwd = Print Working Directory. It shows us the current directory.

1.4 Sort the contents of the Namibia file into alphabetical order and save it as Mzansi.

Command:

`"sort Wildlife/Namibia > Mzansi"`



```
ameer@ameer-VirtualBox: ~$ mkdir Wildlife
ameer@ameer-VirtualBox: ~$ ls
Desktop Documents Downloads Music Pictures Public snap Templates Videos Wildlife
ameer@ameer-VirtualBox: ~$ cd Wildlife
ameer@ameer-VirtualBox: ~/Wildlife$ nano Namibia
ameer@ameer-VirtualBox: ~/Wildlife$ cd ..
ameer@ameer-VirtualBox: ~$ pwd
/home/ameer
ameer@ameer-VirtualBox: ~$ sort Wildlife/Namibia > Mzansi
sort: cannot read: Namibia: No such file or directory
ameer@ameer-VirtualBox: ~$ sort Wildlife/Namibia > Mzansi
ameer@ameer-VirtualBox: ~$ ls
Desktop Documents Downloads Music Mzansi Pictures Public snap Templates Videos Wildlife
ameer@ameer-VirtualBox: ~$
```

Note: Verifying that the text inside of Mzansi text file is in alphabetic order.



```
ameer@ameer-VirtualBox: ~$ cat Mzansi
Kudu
Oribi
Oryx
Puku
Sitatunga
Springbok
Strandjui
Topi
ameer@ameer-VirtualBox: ~$
```

1.5 Move the Namibia file to a new file called Backup in the home folder.

Command:

“mv Wildlife/Namibia Backup”

```
aneer@aneer-VirtualBox:~$ mv Wildlife/Namibia Backup
aneer@aneer-VirtualBox:~$ ls
Backup Desktop Documents Downloads Music Mzansi Pictures Public snap Templates Videos Wildlife
aneer@aneer-VirtualBox:~$
```

Note: Verifying Mzansi text file is inside of backup.

```
aneer@aneer-VirtualBox:~$ mv Wildlife/Namibia Backup
aneer@aneer-VirtualBox:~$ ls
Backup Desktop Documents Downloads Music Mzansi Pictures Public snap Templates Videos Wildlife
aneer@aneer-VirtualBox:~$ cd Backup
bash: cd: Backup: Not a directory
aneer@aneer-VirtualBox:~$ cat Backup
Dryx
Springbok
Strandjuit
Puku
Sitatunga
Topi
Dribi
Kudu
aneer@aneer-VirtualBox:~$
```

1.6 Change the permissions for the Backup file.

1) Firstly, I gave my user account executable permissions.

I made use of the commands:

“sudo chmod u+x Backup” & “sudo chmod u+x Mzansi”

```
aneer@aneer-VirtualBox:~$ sudo chmod 700 /Backup/Mzansi
chmod: cannot access '/Backup/Mzansi': No such file or directory
aneer@aneer-VirtualBox:~$ sudo chmod 700 /Backup/Mzansi
chmod: cannot access '/Backup/Mzansi': No such file or directory
aneer@aneer-VirtualBox:~$ chmod 700 Backup/Mzansi
chmod: command not found
aneer@aneer-VirtualBox:~$ chmod 700 Backup/Mzansi
chmod: cannot access 'Backup/Mzansi': Not a directory
aneer@aneer-VirtualBox:~$ ls -l Backup
-rw-rw-r-- 1 aneer aneer 57 May 24 15:51 Backup
aneer@aneer-VirtualBox:~$ ls -l Mzansi
-rw-rw-r-- 1 aneer aneer 57 May 24 15:54 Mzansi
aneer@aneer-VirtualBox:~$ sudo chmod u+x Backup
aneer@aneer-VirtualBox:~$ ls -l Backup
-rwxrwxr-- 1 aneer aneer 57 May 24 15:51 Backup
aneer@aneer-VirtualBox:~$
```

Note: To verify the updated permissions:

“ls -l Backup” & “ls -l Mzansi”

2) Enter the command:

“chmod 700 ~/Backup/Namibia”

```
ameer@ameer-VirtualBox:~$ sudo chmod u+x Backup
ameer@ameer-VirtualBox:~$ ls -l Backup
-rwxr-xr-x 1 ameer ameer 57 May 24 15:51 Backup
ameer@ameer-VirtualBox:~$ chmod 788 Backup/Mzansi
chmod: cannot access 'Backup/Mzansi': Not a directory
ameer@ameer-VirtualBox:~$ chmod 700 Backup
ameer@ameer-VirtualBox:~$ ls
Backup Desktop Documents Downloads Music Mzansi Pictures PUBLIC tmp tmplost Videos
ameer@ameer-VirtualBox:~$ cd Backup
bash: cd: Backup: Not a directory
ameer@ameer-VirtualBox:~$
```

[Sub Total: 50 Marks]

Question 2: Troubleshooting Video



ITLXA0-22 -
Formative Assessment

Question 3 Scripting

3.1 Information to Log During Backup Operations

Key information to log would include:

- **Timestamp** which function is to record the date and time of the backup operation.
- **Backup Source** specifies the directories or files being backed up.
- **Backup Destination** is there to dictate where the backup is stored.
- **Status** logs whether the backup was successful or failed.
- **Error Alerts** capture any errors encountered during the backup process.

This can be demonstrated within a Virtual machine.

Steps:

- 1) Open your terminal inside of VM1 “Linux2030”.
- 2) Create a backup script (e.g., backup.sh):

“sudo nano ~/backup.sh”

- 3) Add the following logging commands to your script:

“#!/bin/bash

```
LOGFILE=~/.backup.log
```

```
TIMESTAMP=$(date -d "2025-05-25 15:22:00" +"%Y-%m-%d %H:%M:%S")
```

```
SOURCE_DIR=~/.Documents
```

```
DEST_DIR=~/.Backup
```

```
echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >> $LOGFILE
```

```
cp -r $SOURCE_DIR $DEST_DIR
```

```
if [ $? -eq 0 ]; then
```

```
    echo "[$TIMESTAMP] Backup successful" >> $LOGFILE
```

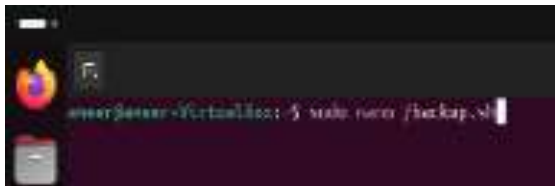
```
else
```

```
    echo "[$TIMESTAMP] Backup failed" >> $LOGFILE
```

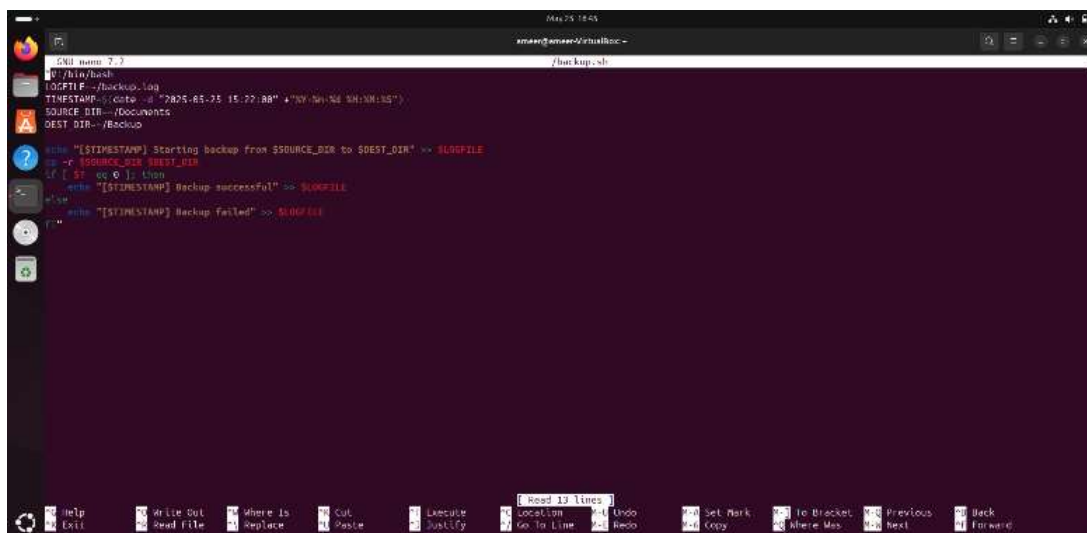
```
fi"
```

Screenshot:

1) Creating the backup script:



2) Insert the script to backup the data and logging whether it succeeds or fails:



3.2 Ensuring Redundancy in the Backup System

Redundancy Strategies include:

Having Multiple Backup Locations that store backups in different physical or cloud locations. Incorporate versioning which relates to keeping multiple versions of backups to recover from different points in time.

Here are the steps:

- 1) Modify your backup script to include a second backup location:

The command is:

```
"SECOND_DEST_DIR=~/.Backup2  
cp -r $SOURCE_DIR $SECOND_DEST_DIR"
```

Written with the previous script, it'll look like this:

```
#!/bin/bash  
LOGFILE=~/.backup.log  
TIMESTAMP=$(date -d "2025-05-25 15:20:00" +"%Y-%m-%d %H:%M:%S")  
SOURCE_DIR=~/.Documents  
DEST_DIR=~/.Backup  
SECOND_DEST_DIR=~/.Backup2  
  
echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >>  
$LOGFILE  
cp -r $SOURCE_DIR $DEST_DIR  
if [ $? -eq 0 ]; then  
    echo "[$TIMESTAMP] Backup to $DEST_DIR successful" >> $LOGFILE  
else  
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" >> $LOGFILE  
fi
```

```

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to
$SECOND_DEST_DIR" >> $LOGFILE

cp -r $SOURCE_DIR $SECOND_DEST_DIR

if [ $? -eq 0 ]; then

    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR successful" >>
$LOGFILE

else

    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" >> $LOGFILE

fi"

```

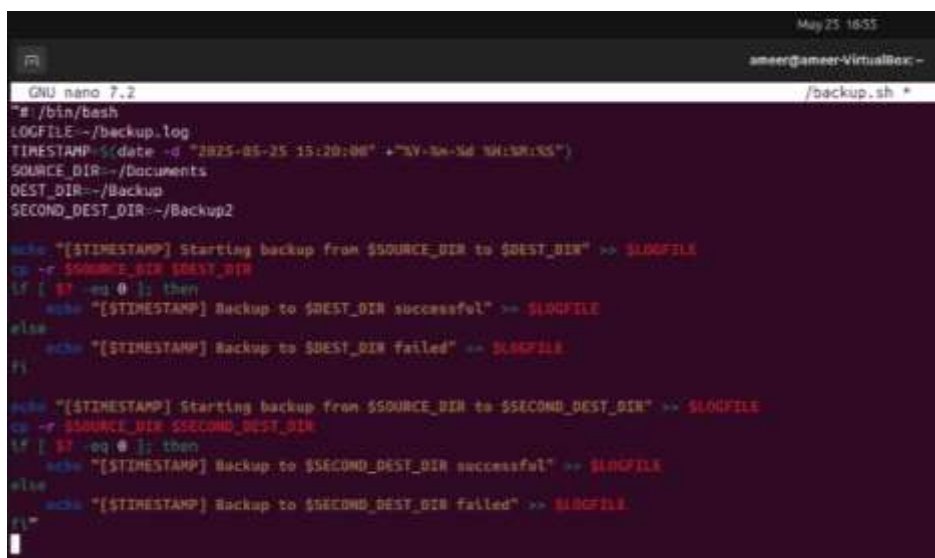
Note: The new variable “SECOND_DEST_DIR=~/.Backup2” should be declared alongside the first destination directory.

Then the copy command “cp -r \$SOURCE_DIR \$SECOND_DEST_DIR” should be added after the first copy.

The script's ‘if’ conditions should be duplicated for the second copy operation.

Screenshot:

- 1) Add the second source destination command to the previous backup script:



```

May 25 16:55
ameer@ameer-VirtualBox: ~
/backup.sh *
GNU nano 7.2
#!/bin/bash
LOGFILE=/backup.log
TIMESTAMP=$(date -d "$(date +%Y-%m-%d %H:%M:%S)")
SOURCE_DIR=/Documents
DEST_DIR=/Backup
SECOND_DEST_DIR=/Backup2

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" >> $LOGFILE
fi

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $SECOND_DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $SECOND_DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" >> $LOGFILE
fi

```

3.3 Mechanism to Alert IT Team in Case of Backup Failures

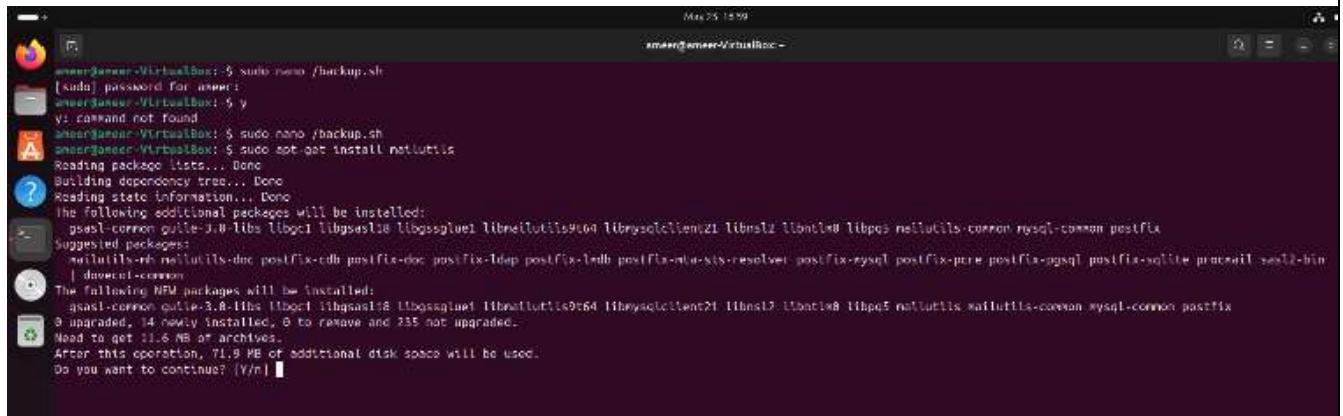
Alert Mechanism:

I made use of an email mechanism to alert the IT team.

These are the steps I followed:

- 1) Install mailutils to send email notifications:

“sudo apt-get install mailutils”



```
ameer@ameer-VirtualBox: ~$ sudo nano /backup.sh
[sudo] password for ameer:
ameer@ameer-VirtualBox: ~$ y
y: command not found
ameer@ameer-VirtualBox: ~$ sudo nano /backup.sh
ameer@ameer-VirtualBox: ~$ sudo apt-get install mailutils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gssapi-common guile-3.0 libs libgc1 libgssapi16 libgssglue1 libmailutils9t64 libmysqlclient21 libnsl2 libntlm0 libpopt0 mailutils-common mysql-common postfix
Suggested packages:
  mailutils-rb mailutils-doc postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sys-resolver postfix-mysql postfix-pcre postfix-pgsql postfix-sqlite procmail sendmail-bin
  dovecot-common
The following NEW packages will be installed:
  gssapi-common guile-3.0-libs libgc1 libgssapi16 libgssglue1 libmailutils9t64 libmysqlclient21 libnsl2 libntlm0 libpopt0 mailutils mailutils-common mysql-common postfix
0 upgraded, 14 newly installed, 0 to remove and 235 not upgraded.
Need to get 11.6 MB of archives.
After this operation, 71.9 MB of additional disk space will be used.
Do you want to continue? (Y/n) Y
```

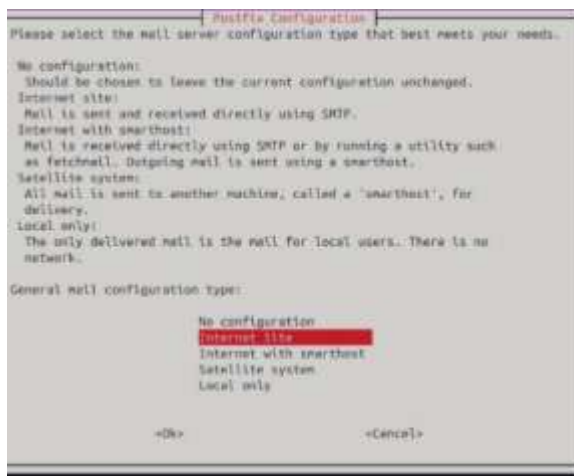
Note: After confirming the command, you’ll be prompted to agree with ‘y’ or ‘n’ to allow the installation. Select y.

You’ll be greeted by the mailutils application. Follow the prompts, by selecting the Internet site option.

Lastly, you’ll be asked to provide a suffix for your email address.

Use: ‘@gmail.com’

Allow the mailutils installation to complete.



1) Modify your script to send an email on failure:

```
“if [ $? -ne 0 ]; then  
    echo "Backup failed" | mail -s "Backup Alert" it-team@example.com  
fi”
```

It'll look like this:

```
#!/bin/bash  
  
LOGFILE=~/.backup.log  
  
TIMESTAMP=$(date -d "2025-05-25 15:20:00" +"%Y-%m-%d %H:%M:%S")  
  
SOURCE_DIR=~/.Documents  
  
DEST_DIR=~/.Backup  
  
SECOND_DEST_DIR=~/.Backup2  
  
  
echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >>  
$LOGFILE  
  
cp -r $SOURCE_DIR $DEST_DIR  
  
if [ $? -eq 0 ]; then  
    echo "[$TIMESTAMP] Backup to $DEST_DIR successful" >> $LOGFILE  
else  
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" >> $LOGFILE  
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" | mail -s "Backup Alert" it-  
team@example.com  
fi  
  
  
echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to  
$SECOND_DEST_DIR" >> $LOGFILE  
  
cp -r $SOURCE_DIR $SECOND_DEST_DIR  
  
if [ $? -eq 0 ]; then  
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR successful" >>  
$LOGFILE
```

else

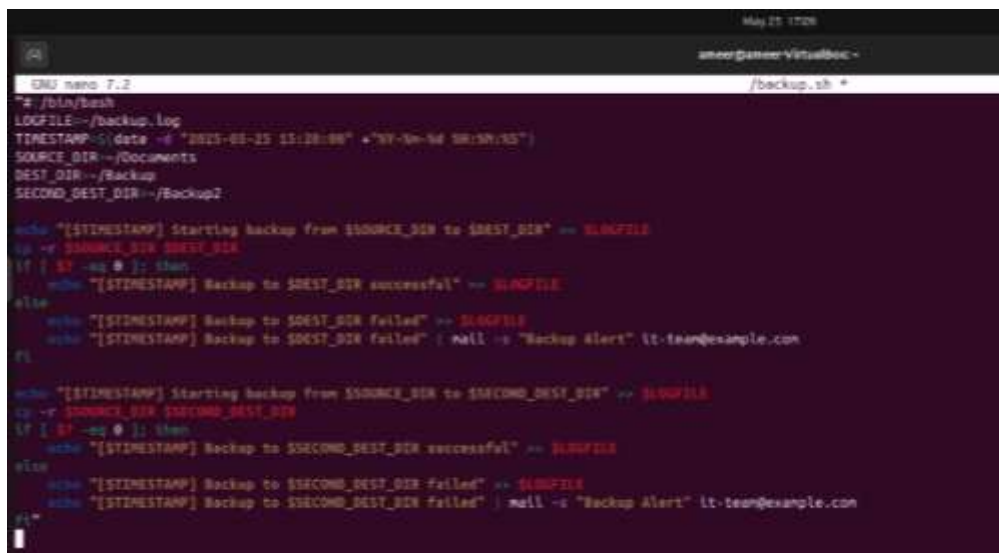
echo "[\$TIMESTAMP] Backup to \$SECOND_DEST_DIR failed" >> \$LOGFILE

echo "[\$TIMESTAMP] Backup to \$SECOND_DEST_DIR failed" | mail -s "Backup Alert" it-team@example.com

fi"

Note: The email body should include useful diagnostic information like the timestamp, destination directory and failure message to help the team quickly understand the issue.

Screenshot:



```
#!/bin/bash
LOGFILE=/backup.log
TIMESTAMP=$(date +%Y-%m-%d %H:%M:%S)
SOURCE_DIR=/Documents
DEST_DIR=/Backup
SECOND_DEST_DIR=/Backup2

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" >> $LOGFILE
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" | mail -s "Backup Alert" it-team@example.com
fi

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $SECOND_DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $SECOND_DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" >> $LOGFILE
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" | mail -s "Backup Alert" it-team@example.com
fi
```

3.4 Ensuring Team Members Can Understand and Manage the Backup System

Documentation Strategies I implemented:

Adding comments in the script to explain key parts. I also made a separate document explaining how to run and manage the backup script.

Steps I followed:

- 1) Add comments to your script:

"# This script backs up the Documents directory to two locations It sends an email to the IT team when backups fail."

```

"# This script backs up the Documents directory to two locations It sends an email to the IT team when backups fail."
#!/bin/bash
LOGFILE=~/.backup.log
TIMESTAMP=$(date -d "2025-05-25 15:20:00" +"%Y-%m-%d %H:%M:%S")
SOURCE_DIR=~/.Documents
DEST_DIR=~/.Backup
SECOND_DEST_DIR=~/.Backup2

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" >> $LOGFILE
    echo "[$TIMESTAMP] Backup to $DEST_DIR failed" | mail -s "Backup Alert" it-team@example.com
fi

echo "[$TIMESTAMP] Starting backup from $SOURCE_DIR to $SECOND_DEST_DIR" >> $LOGFILE
cp -r $SOURCE_DIR $SECOND_DEST_DIR
if [ $? -eq 0 ]; then
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR successful" >> $LOGFILE
else
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" >> $LOGFILE
    echo "[$TIMESTAMP] Backup to $SECOND_DEST_DIR failed" | mail -s "Backup Alert" it-team@example.com
fi

```

2) Create a user manual (e.g., backup_manual.txt):

`"nano ~/.backup_manual.txt"`

3) Include instructions on how to run the script and what to do in case of errors.

backup_manual.txt:

"Backup Script User Manual"

Introduction

This manual provides instructions for running the backup script and troubleshooting common issues.

How the Script Works

- **Log File:** Records the backup process in `~/.backup.log`.
- **Source Directory:** Backs up files from `~/.Documents`.
- **Destinations:** Copies files to `~/.Backup` and `~/.Backup2`.
- **Error Alerts:** Logs success or failure and sends an email alert on failure.

How to Run the Backup Script

1. Open your terminal.

2. Ensure the script is executable:

```
"chmod +x backup_script.sh"
```

3. Run the script:

```
"./backup_script.sh"
```

What to Expect

- Log entries in `~/backup.log` for every attempt.
- Email alerts for any failures.

Error Alerting

1) **Check the Log:** Review `~/backup.log` for error messages.

2) Common Issues:

- **Permission Denied:** Use `sudo` if needed.
- **Directory Not Found:** Ensure `~/Documents`, `~/Backup`, and `~/Backup2` exist.
- **Email Issues:** Verify mail configuration.

3) **Retry:** After fixing issues, rerun the script.

4) **Seek Help:** Contact IT support with error details if problems persist.

Additional Tips

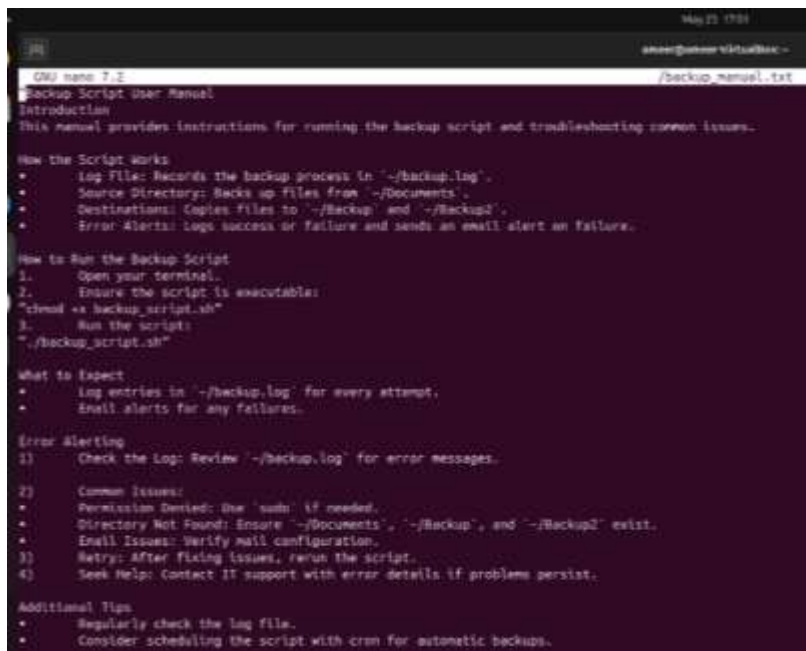
- Regularly check the log file.
- Consider scheduling the script with cron for automatic backups.

Contact for Support

For further assistance, contact your system administrator or IT support team.

End of Manual

Screenshot:



```
May 21 17:01
samer@ubuntu-virtualbox: ~
GNU nano 7.2
Backup Script User Manual
Introduction
This manual provides instructions for running the backup script and troubleshooting common issues.

How the Script Works
• Log File: Records the backup process in './backup.log'.
• Source Directory: Backs up files from './Documents'.
• Destinations: Copies files to './Backup1' and './Backup2'.
• Error Alerts: Logs success or failure and sends an email alert on failure.

How to Run the Backup Script
1. Open your terminal.
2. Ensure the script is executable:
   chmod +x backup_script.sh
3. Run the script:
   ./backup_script.sh

What to Expect
• Log entries in './backup.log' for every attempt.
• Email alerts for any failures.

Error Alerting
1) Check the Log: Review './backup.log' for error messages.
2) Common Issues:
   • Permission Denied: Use 'sudo' if needed.
   • Directory Not Found: Ensure './Documents', './Backup1', and './Backup2' exist.
   • Email Issues: Verify mail configuration.
3) Retry: After fixing issues, rerun the script.
4) Seek Help: Contact IT support with error details if problems persist.

Additional Tips
• Regularly check the log file.
• Consider scheduling the script with cron for automatic backups.
```

3.5 Importance of Regular Maintenance of the Backup Script

Reasons for Regular Maintenance:

- 1) **Adapt to Changes:** Ensures the script adapts to changes in file structure or backup requirements.
- 2) **Fix Bugs:** Regularly checks for and make sure to fix any bugs or issues that appear.
- 3) **Update Security:** Ensures that the script follows the latest security practices.

Practical ideas can include scheduling regular reviews of the script (monthly). Activities such as script reviews and updates for security patches should be done. Remember to document any changes made during maintenance in a changelog, using the “#” followed by your comment.

[Sub Total 10 Marks]

Reference List:

Question 1:

<https://www.virtualbox.org/manual/>

<https://youtu.be/rJ9ysibH768>

<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>

https://youtu.be/DLeATFgGM-A?si=E2WQAWNv9BNRjr_h

<https://youtu.be/LnKoncbQBsM?si=BvC0fHzDSCPjoZj4>

<https://help.ubuntu.com/>

Question 2:

<https://tldp.org/LDP/tlk/kernel/kernel.html>

<https://youtu.be/Xi7ACzmzk8k?si=iq7qHje5nOgHjjS9>

<https://youtu.be/iypmo5XsfWw?si=yQpxHXatzRpGrjSH>

https://youtu.be/UGFtnLaVo80?si=Ajv3vx_EWDwix7Lq

<https://youtu.be/8xFAVvrfEdg?si=mtnxuAEaTL7qO4Ug>

<https://youtu.be/HYiTJtYmmzg?si=enuLGQUC9YdY-0hw>

<https://www.linuxfromscratch.org/blfs/view/stable-systemd/postlfs/initramfs.html>

https://www.reddit.com/r/linux4noobs/comments/10ab5ia/confused_about_purpose_of_initramfs_during_boot/

<https://last9.io/blog/system-logs/>

<https://access.redhat.com/solutions/1958>

<https://www.kernel.org/doc/html/v4.14/admin-guide/kernel-parameters.html>

<https://www.linuxquestions.org/>

Question 3:

<https://www.connectwise.com/blog/backup-strategy-best-practices>

<https://www.rapid7.com/blog/post/2016/07/13/backup-log-checks-and-what-they-can-tell-you/>

<https://clouglobal.com/data-protection-2-0-why-backups-and-redundancy-are-non-negotiable/>

<https://docs.aws.amazon.com/prescriptive-guidance/latest/security-best-practices/monitor-alert.html>

https://www.reddit.com/r/Unity3D/comments/16stllk/good_practice_for_commenting_in_scripts/

https://youtu.be/qNQ_p4jr3A?si=vvk6G-p60qtssjoR

https://youtu.be/_iQcEnaOgcs?si=KlBloYFxUkAQDSPE

<https://www.youtube.com/watch?v=ElEw7cA2Ea8>

<https://www.youtube.com/watch?v=tK9Oc6AEnR4&pp=0gcJCdgAo7VqN5tD>

<https://discourse.ubuntu.com/t/basic-backup-shell-script/36419>

<https://www.redhat.com/en/blog/linux-monitoring-and-alerting>
